

01e — DNS and How It Works

Key Concepts

Term	Definition
DNS	System that translates domain names to IP addresses
Recursive Resolver	Server that does the full lookup on your behalf (ISP / 8.8.8.8)
Root Name Server	Top of the hierarchy — knows where each TLD server is
TLD Server	Knows which authoritative server handles a domain (e.g. all .com domains)
Authoritative Server	Has the final answer — the actual IP for a domain
TTL (Time To Live)	How long a resolver may cache a DNS record before re-asking
A Record	Maps domain → IPv4 address
AAAA Record	Maps domain → IPv6 address
CNAME	Alias — points to another domain name (not an IP)
MX Record	Specifies mail servers for a domain
NS Record	Lists the authoritative name servers for a domain
TXT Record	Arbitrary text — used for domain verification, SPF, DKIM

How It Works

Full DNS resolution chain (cache-miss scenario):

1. Browser checks its own DNS cache
2. OS checks its local DNS cache
3. Request goes to **Recursive Resolver** (ISP or public e.g. 8.8.8.8)
4. Resolver asks **Root Name Server** → returns address of TLD server
5. Resolver asks **TLD Server** (e.g. **.com**) → returns address of authoritative server
6. Resolver asks **Authoritative Server** → returns the actual IP
7. Resolver caches the result (per TTL) and returns IP to browser
8. Browser connects to the IP

The browser only talks to the recursive resolver. The resolver does all the legwork.

TTL behaviour:

- High TTL (86400s) → cached for 1 day, low DNS traffic
 - Low TTL (60s) → expires fast, changes propagate quickly, higher DNS load
 - Lowering TTL takes effect only after the current TTL expires
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DNS Record Types

Record	Maps	Common Use
A	domain → IPv4	Most web servers
AAAA	domain → IPv6	Modern dual-stack servers
CNAME	domain → domain	www.site.com → site.com
MX	domain → mail server	Email routing
TXT	domain → text	SPF, DKIM, domain ownership
NS	domain → name server	Delegating a zone

CNAME vs A Record:

- A record: points directly to an IP
- CNAME: points to another domain name which resolves to an IP
- If IP changes, update one A record — all CNAMEs follow automatically

Industry Examples

Company	DNS Usage
Netflix	DNS-based geo-routing — returns different IPs based on user location, directing users to nearest CDN node
AWS Route 53	Latency-based and geolocation routing; health-check failover via DNS
Cloudflare (1.1.1.1)	Fast public recursive resolver; also authoritative DNS for millions of domains
Google (8.8.8.8)	Public recursive resolver, bypasses slow ISP DNS

Zero-Downtime Migration Pattern

Scenario: Moving `api.yourapp.com` to a new server IP

1. Check current TTL on A record (e.g. 86400 = 1 day)
2. **Lower TTL to 60s** — wait the full current TTL duration (24 hrs) for all caches to expire
3. Verify new server is live and healthy
4. **Update A record** to new IP
5. Within 60s, all clients re-resolve and hit new server
6. **Raise TTL back** to 86400 once migration is stable

Interview Questions

Q: What is DNS and why do we need it? A: DNS translates human-readable domain names to IP addresses. Without it, users would need to memorize IPs for every site.

Q: Walk me through a full DNS lookup for github.com. A: Browser cache → OS cache → recursive resolver → root server → TLD server → authoritative server. Each layer caches the result per TTL.

Q: What is TTL and why does it matter? A: TTL controls how long resolvers cache a DNS record. Low TTL = fast propagation of changes. High TTL = less DNS traffic. Critical for planning IP migrations.

Q: What's the difference between an A record and a CNAME? A: A record maps a domain directly to an IPv4 address. CNAME maps a domain to another domain name. CNAMEs are useful so multiple aliases track one A record automatically.

Q: How does Netflix route users to the nearest server using DNS? A: Netflix's authoritative DNS server inspects where the resolver request comes from and returns a different IP based on geography, directing users to the nearest CDN node.